

BS Software Engineering Fall Afternoon - B

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Subject : Software Engineering

Software System Analysis: Banking System

A. Purpose

A Banking System is a software platform that enables financial institutions to manage customer accounts, process transactions, and deliver banking services securely and efficiently. Its core purpose is to automate financial operations — deposits, withdrawals, transfers, loan processing, and account management — while ensuring accuracy, regulatory compliance, and data security across all operations.

B. Users

- **Retail customers** – individuals managing personal accounts, transfers, and payments
- **Corporate/business clients** – companies managing payroll, bulk transactions, and trade finance
- **Bank tellers/staff** – employees performing in-branch transactions
- **Bank administrators** – manage system configuration, user roles, and permissions
- **Loan officers** – process and approve loan applications
- **Auditors/compliance officers** – review transactions for regulatory adherence

C. Stakeholders

- Bank management and executives
- Shareholders/investors
- Regulatory bodies (e.g., central banks, financial authorities)
- IT/software development and support teams

- Third-party service providers (payment gateways, credit bureaus)
- End customers (retail and corporate)

D. Functional Requirements (5)

1. The system shall allow users to create, view, update, and close bank accounts.
2. The system shall process deposits, withdrawals, and fund transfers between accounts.
3. The system shall generate account statements and transaction histories.
4. The system shall support loan application submission, approval workflows, and repayment tracking.
5. The system shall authenticate users via secure login (e.g., multi-factor authentication) before granting access.

E. Non-Functional Requirements (5)

1. **Security** – All transactions and data must be encrypted (e.g., AES-256, TLS) both in transit and at rest.
2. **Availability** – The system shall maintain 99.9% uptime to support 24/7 banking operations.
3. **Performance** – Transaction processing shall complete within 2–3 seconds under normal load.
4. **Scalability** – The system shall support a growing number of concurrent users without performance degradation.
5. **Regulatory Compliance** – The system shall comply with financial regulations such as PCI-DSS, KYC, and AML standards.

F. Risks (3)

1. **Cybersecurity threats** – Data breaches, phishing, or fraud attacks could compromise sensitive financial data.
2. **System downtime** – Outages could disrupt critical banking services, causing financial and reputational loss.
3. **Regulatory non-compliance** – Failure to meet evolving financial regulations could result in legal penalties.

G. Quality Attributes (3)

1. **Reliability** – The system must consistently perform accurate transactions without failure.
2. **Security** – Strong protection of sensitive financial and personal data is essential.
3. **Usability** – The interface must be intuitive for users across varying levels of technical literacy.

H. Where AI Could Be Used

- **Fraud detection** – AI models can analyze transaction patterns in real time to flag suspicious activity.
- **Credit scoring** – Machine learning can assess loan applicants' creditworthiness more accurately.
- **Chatbots/virtual assistants** – AI-powered support for customer queries and basic transactions.
- **Personalized financial recommendations** – AI can suggest savings plans or investment products based on user behavior.

I. What Could Go Wrong If AI Is Used Incorrectly

- **Biased decision-making** – Poorly trained AI models could unfairly deny loans or flag legitimate transactions based on biased data.
- **False positives/negatives in fraud detection** – Legitimate transactions could be blocked, or actual fraud could go undetected, damaging trust.
- **Lack of transparency (black-box decisions)** – Customers and regulators may be unable to understand why an AI made a specific decision, raising accountability and compliance concerns.